

# UV254 COD Sensor User Manual

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Xi'an Desun Uniwill Electronic Technology Co., Ltd.

**Please Read This Manual Carefully Before Use.**





## Preface

Thank you very much for choosing our company!

Please read this manual carefully before using this product, and save it for reference. Please observe the operating procedures and precautions in this manual.

To ensure that the after-sale protection provided by this instrument is effective, please do not use and maintain the instrument in any way other than specified in this manual.

As the failure to observe the precautions specified in this operation manual, any failures and losses caused are not covered by the manufacturer's warranty, and the manufacturer does not assume any related responsibility. Please keep all documents in a safe place. If in doubt, please contact our after-sales service department.

When receiving the instrument, please carefully open the package and check whether the instrument and accessories are damaged due to shipping. If damage is found, please contact Desun Uniwill after-sales service department and keep the packaging for return.

When the instrument fails, please do not repair it by yourself, please contact our after- sales service department.

# UV254 COD Sensor User Manual

## I. Overview

DS500 series COD sensor is a new generation of environmental protection type COD sensor launched by our company, which is reagent free, pollution-free, more economic and environmental protection. Small size, more convenient installation, online continuous water quality monitoring. Automatic compensation for turbidity interference, with automatic cleaning device, even long-term monitoring still has excellent stability.

Many organic substances dissolved in water absorb ultraviolet light. Therefore, the total amount of organic pollutants in water can be measured by measuring the absorption of these organic substances by ultraviolet light at a wavelength of 254 nm.

DS500 series sensors use two light sources, one 254nm ultraviolet light and one 850nm infrared light, which can automatically compensate the optical path attenuation and turbidity effects, thus achieving more stable and reliable measurement values.

### Features:

- ✧ Digital sensor, RS-485 output, support MODBUS
- ✧ No reagents, no pollution, more economical and environmentally friendly
- ✧ Automatic compensation for turbidity interference for excellent test performance
- ✧ Can Test COD/BOD/TSS at same time
- ✧ Built-in Temperature

### Technical parameter:

| Item                | Parameters                                                  |                                                              |
|---------------------|-------------------------------------------------------------|--------------------------------------------------------------|
| Model               | Measuring Range                                             | Application Field                                            |
| DS500<br>(6 mm gap) | COD 0 ~ 500 mg/L<br>TSS 0~600 mg/L<br>BOD 0~300 mg / L      | Tap water                                                    |
|                     | COD 0 ~ 2000 mg/L<br>TSS 0~1000 mg/L<br>BOD 0~1000 mg / L   | ① Sewage treatment plant outlet<br>② ETP<br>③ Natural waters |
| DS501<br>(1 mm gap) | COD 6000~10000 mg/L<br>TSS 0~5000 mg/L<br>BOD 0~5000 mg / L | Industrial process control                                   |

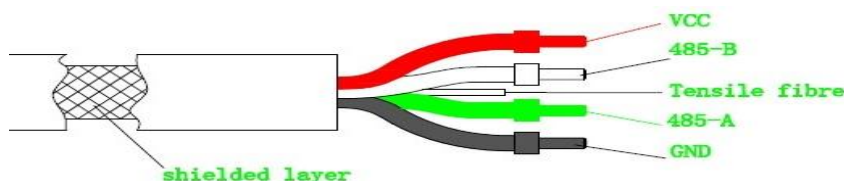


|                     |                                                         |
|---------------------|---------------------------------------------------------|
| Power supply        | 12VDC+/-5%                                              |
| Principle           | UV 254                                                  |
| Output signal       | RS485/Modbus                                            |
| Accuracy            | 0.01mg/L                                                |
| Calibration         | 1 or 2 point calibration                                |
| Housing material    | SS316                                                   |
| Self-cleaning Brush | Have/ No<br>(When the COD range is >2000mg/L, no brush) |

## II. Cable Information for Sensors

4 wire AWG-24 or AWG-26 shielding wire. OD=5mm

- Red line----power supply (VCC)
- White line---- 485 data -B (485-B)
- Yellow line---- 485 data -A (485-A)
- Black line----ground (GND)
- Bare line---- shielding layer



The power supply must be DC 6-12V +/-5%, current <50mA.

## III. Sensor Installation

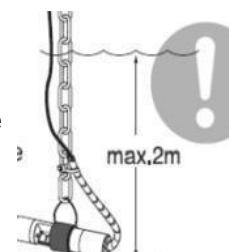
Spread the cables before handling.

1. When the sensor is suspended, the sensor caused by water flow should be prevented from hitting the wall or other water conservancy facilities.

If the water flow is very urgent, please fix the sensor

2. When installing the sensor, the depth from the water surface should not exceed 2 meters.

3. Considering the fluctuation of water level, submerge the sensor below the lowest possible water level of 30CM.

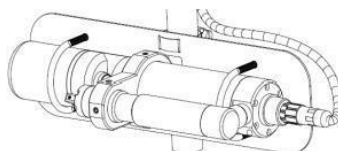


4. The sensor is placed in a position where there are no bubbles in the water.

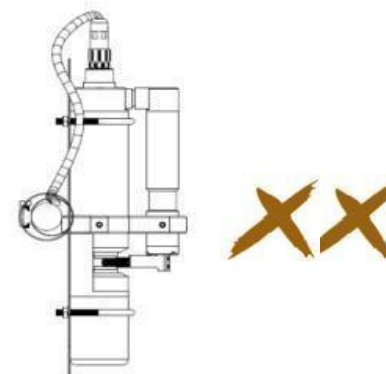
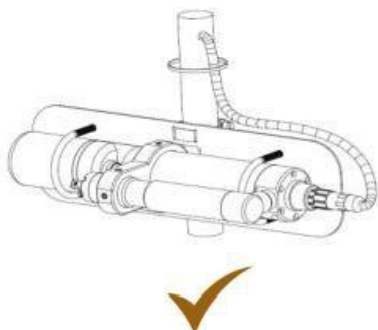


5. Install a cable cover on the outside of the sensor cable.

6. Fix the sensor as shown below



7. Fixed firmly. The sensor needs to be placed horizontally.



## Warning

1. Please install the cable cover correctly. Otherwise, the cable may be damaged due to damage during maintenance.
2. Do not use the sensor cable to lift the sensor.
3. Do not cover the measuring surface with the lifting attachment.

## IV. COD Sensor Calibration

- DSC500 UV254 COD sensor software (see modbus user manual for details).
- Provides 1- or 2-point COD calibration options in mg/L. KHP (potassium hydrogen phthalate, C<sub>8</sub>H<sub>5</sub>KO<sub>4</sub>), CAS# 877-24-7, is a commonly used stain for environmental studies and can be used for BOD calibration.
- The conversion formula for BOD<sub>254nm</sub> as follows:  $c(\text{COD}) = 1.176 * c(\text{KHP})$

## Preparation

1. Accurately weigh 1.2754 g of KHP into a 1000 mL flask.
2. Distilled or deionized water and filled until the highest scale. This solution contains 1500mg COD.
3. Take 100 mL of this solution into a 1000 mL flask and fill it to the highest mark with distilled or deionized water. Shake well to ensure a COD content of 150 mg/L.
4. Store the concentrated standard solution in a black glass bottle and store at low temperature to prevent decomposition.
5. The diluted standard needs to be used within 24 hours of preparation.

**Note:** The specific choice of standard solution is based on the actual range of the sensor.



## Calibration (2-point calibration)

Step 1: Select the theoretical value according to the range, e.g. if the range is 100mg/l, then choose 12 for theoretical value 1; choose 84 for theoretical value 2.

Step 2: Before starting this, Reset  $K=1, B=0$ .

Put the sensor into the test solution of 12mg/l, wait for the test data to stabilize, then enter the average value of the measured value as the measured value 1.

Place the sensor in another test solution of 84mg/l, wait for the test data to stabilize and enter the average of the measured values as the measured value 2.

Step 3: Click **OK** via the calibration help software, you will get the new K,B values, save the new K,B values in the testing software.

COD calibration is completed.

## V. Maintenance Schedule and Method

### Maintenance schedule

Cleaning is important to ensure accurate measurements.

| Maintenance Work                           | Maintenance Frequency                        |
|--------------------------------------------|----------------------------------------------|
| Calibration (if required by the agent)     | Calibrate according to the required schedule |
| Maintain and check the self-cleaning brush | Every 18 months (see wiper doc for details)  |

### Maintenance Methods&Routine maintenance

**1) Surface:** Flush the outer surface of the sensor. If any dirt remains, wipe it with a soft, damp cloth. For some dirt that is difficult to clean, add household detergent to the water and rinse it clean. .

**2) Check the cable:** Do not tighten the cable during normal operation, otherwise the internal conductor will be broken and the sensor will not work properly.

**3) Check the sensor's measurement window for dirt and confirm that the cleaning brush is working properly.**

**4) Check the sensor surface for damage.**

**Note:** There are precision optics and electronics inside the sensor, so please avoid severe mechanical shock.



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**Common Problems:**

| Error                                                                | Possible causes                          | Solutions                           |
|----------------------------------------------------------------------|------------------------------------------|-------------------------------------|
| Unstable reading                                                     | Connection failure                       | Reconnect the controller and cables |
|                                                                      | Cable failure                            | Please contact us                   |
| Measured values are too high, too low, or the values remain unstable | The sensor measurement window is blocked | Cleaning surface                    |

## VI. Quality Assurance

The online COD warranty period is 1 year from the delivery date.

If the product is found to be faulty during the warranty period, our company promises to repair or replace the faulty product free of charge, or return the order for the faulty product to the user after deducting the first shipping fee. Repaired or replaced products will only enjoy the remaining warranty period.

Vulnerable parts are not covered by the warranty, such as consumables (light source, pipes, etc.). During the warranty period, please contact our sales staff or their agents for technical support.

After receiving the product quality problem provided by the customer, our company will confirm whether the product needs repair within 2 weeks; if the repair permit is not obtained, the user must not send the product back without authorization.

## VII. Warranty Limit

The warranty does not cover the following:

- Damage caused by natural disasters, riots, war and other force majeure factors.
- Damage caused by improper use, negligence, accidents or improper use by the user.
- Shipping costs incurred for the return of the product.
- Costs incurred for postage of product parts during the warranty period.
- Travel costs for out-of-warranty repairs.

The remedies such as repair, replacement, or refund of the payment as described above are special cases that do not violate this warranty. The remedies such as replacement or refund of the payment are for our products. Based on strict liability or other legal theories, our company is not responsible for any other damage caused by product defects or negligent operation, including the subsequent damage caused by causality with these situations.